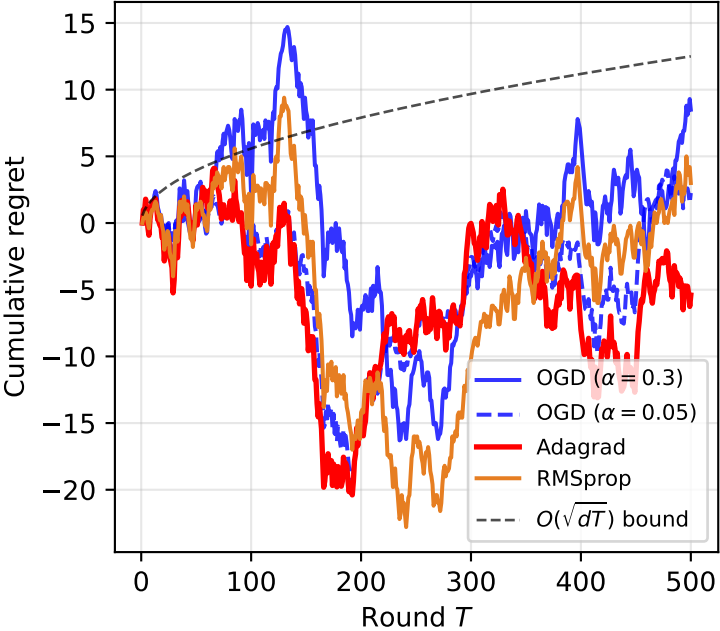
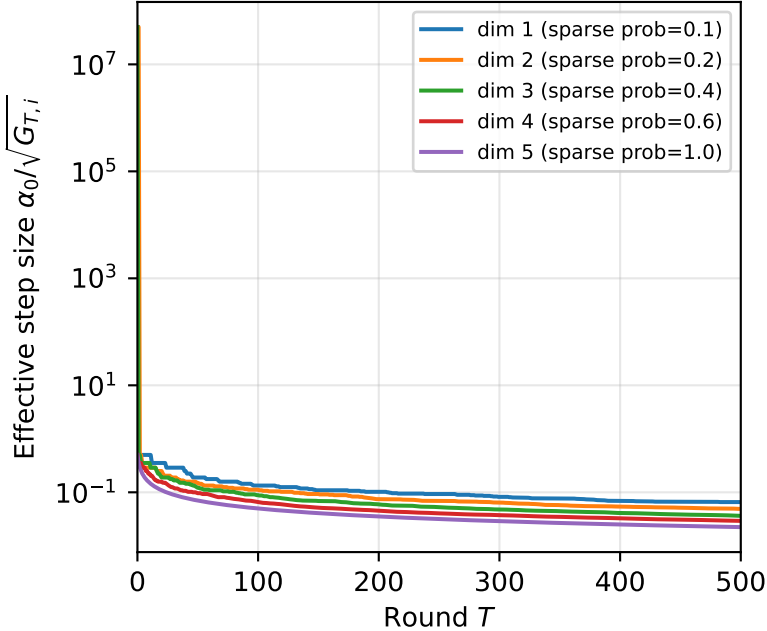


Q1: Adagrad Regret, Heavy Ball Stability, and Theoretical Foundations

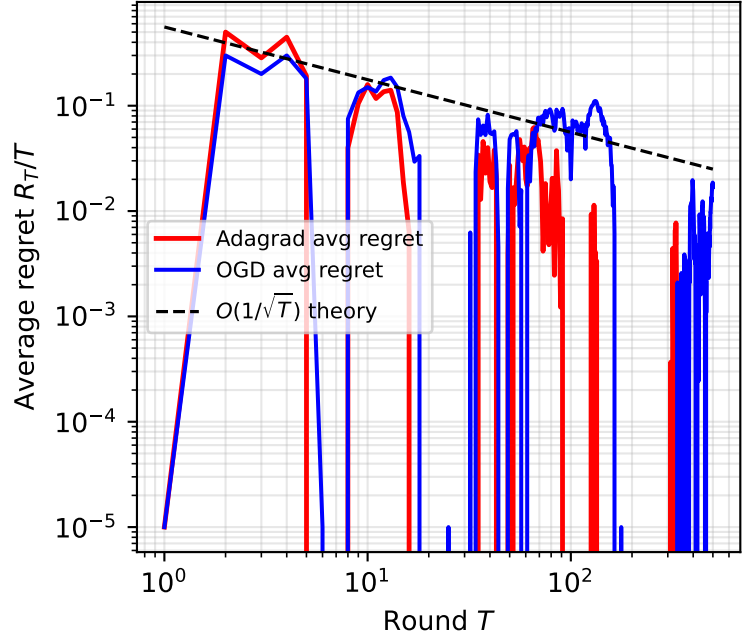
Online Convex Opt.: Regret Comparison



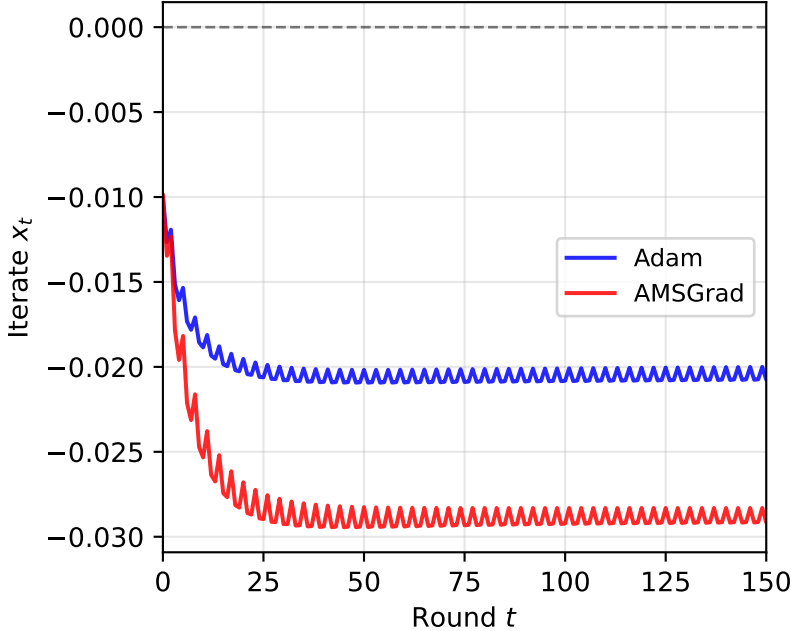
Adagrad Per-Coordinate Step Sizes



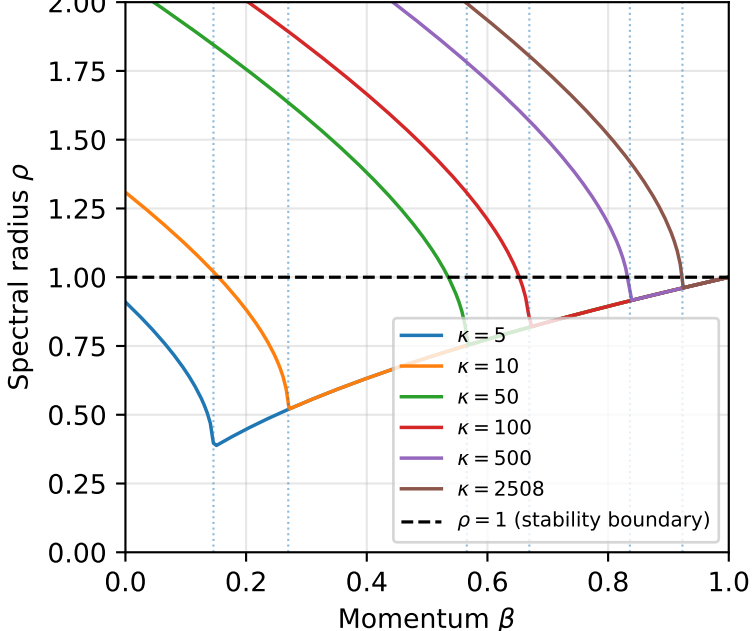
Average Regret: $O(1/\sqrt{T})$
(empirical slope=nan)



Adam vs AMSGrad: Oscillation Example



Heavy Ball Stability: ρ vs β



Polyak-Łojasiewicz Condition

